



Modelling Financial Markets

Seminar 19: Review and Project Workshop — Hands-on



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAGR: Compound Annual Growth Rate
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ ES: Expected Shortfall
- ▣ ETF: Exchange-Traded Fund
- ▣ FHS: Filtered Historical Simulation
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ HS: Historical Simulation
- ▣ WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Outline

Introduction

Part A: Derivations Across the Course

Part B: Integrated Case Study

Part C: The Project Proposal

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Redo the key derivations of the course step by step: returns, GARCH forecasts, VaR and ES, Kupiec, multiple testing
- ▣ Run the full chain on one market: stylised facts, GARCH, VaR 1%, backtest
- ▣ Spot the look-ahead and calendar pitfalls in your own code
- ▣ Write a one-page project proposal

- ▣ Background: Lecture 19 and the lectures it reviews; textbooks [Franke, Härdle & Hafner \(2019\)](#) and [McNeil, Frey & Embrechts \(2015\)](#)

Structure

- ▣ **Part A:** ten short derivations across the course (A1–A10)
- ▣ **Part B:** an integrated case study on BET and Bitcoin, with inference (B1–B8)
- ▣ **Part C:** the project proposal: template, worked example, your proposal (C1); critique of an AI-written backtest (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Sign convention: loss $L = -r$ in % of the position; VaR and ES are positive; the level is the tail probability (VaR 1%, ES 2.5%)
- Code: [seminar notebook \(Colab\)](#), built on the Quantlet MFM_ch19_seminar

Solved and Proposed Problems

Solved: worked examples in this deck

- ▣ Part A: A1 (returns and CAGR), A3 (GARCH forecast), A5 (Normal VaR and ES), A7 (Kupiec test), A9 (multiple testing)
- ▣ Part B: B1 (stylised facts of BET), B3 (GARCH- t for BET), B5 (VaR 1% backtest for BET), B7 (the look-ahead trap)
- ▣ Part C: a worked project proposal

Proposed: you solve them

- ▣ Part A: A2 (model A1), A4 (model A3), A6 (model A5), A8 (model A7), A10 (model A9)
- ▣ Part B: B2 (model B1), B4 (model B3), B6 (model B5), B8 (model B7)
- ▣ Part C: C1, your team's proposal; C2, critique an AI-written backtest (model: B5, B7)

A1. Simple and Log Returns, CAGR [Solved]

Task

- ▣ A price moves $100 \rightarrow 150 \rightarrow 120$ over two years
- ▣ Compute the simple returns R_t and the log returns $r_t = \ln(1 + R_t)$
- ▣ Show that log returns add up over time; compare the mean simple return with the CAGR (compound annual growth rate)

Solution

- ▣ $R_1 = 50.0\%$, $R_2 = -20.0\%$; $r_1 = 0.4055$, $r_2 = -0.2231$
- ▣ $r_1 + r_2 = 0.1823 = \ln(120/100)$: log returns aggregate exactly
 - ▶ Simple returns do not: $R_1 + R_2 \neq 20\%$
- ▣ Mean simple return 15.0% a year, but CAGR $= 1.2^{1/2} - 1 = 9.54\%$
- ▣ The arithmetic mean overstates growth when returns vary (Chapter 1)

A2. Returns over Three Years [Proposed]

Solution discussed in class

Task

- ▣ Model: A1
- ▣ Prices $100 \rightarrow 80 \rightarrow 100 \rightarrow 110$: simple and log returns for each year
- ▣ Mean simple return, CAGR and the sum of log returns; which one describes the investor's experience?

A3. GARCH(1,1) Forecast for BET [Solved]

Task

- BET daily returns in %, GARCH(1,1)- t estimated on 2000–2026: $\omega = 0.0448$, $\alpha = 0.192$, $\beta = 0.799$
- Long-run variance, annualised volatility and half-life of shocks
- Tomorrow's variance is $\sigma_{T+1}^2 = 2.79$ ($\sigma = 1.67\%$): forecast the variance on day 10 and the 10-day volatility

Solution

- $\alpha + \beta = 0.9907$; $\bar{\sigma}^2 = \omega / (1 - \alpha - \beta) = 4.82$, i.e. 2.20% a day, 34.9% a year
- $h_{1/2} = \ln 0.5 / \ln(\alpha + \beta) = 74.3$ days
- $\sigma_{T+10}^2 = \bar{\sigma}^2 + (\alpha + \beta)^9 (\sigma_{T+1}^2 - \bar{\sigma}^2) = 2.95$
 - $(\alpha + \beta)^9 = 0.919$: today is below the long run, so the forecast rises
- 10-day volatility $\sqrt{\sum_{h=1}^{10} \sigma_{T+h}^2} = \sqrt{28.71} = 5.36\%$, above $\sqrt{10} \sigma_{T+1} = 5.28\%$

A4. GARCH Forecast with Given Parameters [Proposed]

Solution discussed in class

Task

- ▣ Model: A3
- ▣ $\omega = 0.02$, $\alpha = 0.10$, $\beta = 0.88$, returns in %; $\sigma_{T+1}^2 = 4$
- ▣ Long-run variance and annualised volatility, half-life, σ_{T+5}^2 and the variance of the 5-day return
- ▣ Is the 5-day volatility above or below $\sqrt{5} \sigma_{T+1}$? Why?



A5. Normal VaR 1% and ES 2.5% [Solved]

Task

- A position of 1 million RON in the BET; daily return $r \sim N(\mu, \sigma^2)$ with $\mu = 0.05\%$, $\sigma = 1.4\%$
- Compute $\text{VaR}_{1\%} = -(\mu + \sigma z_{0.01})$ and $\text{ES}_{2.5\%} = -\mu + \sigma \varphi(z_{0.025})/0.025$, in % and in RON
- Why did Basel replace VaR 1% with ES 2.5%?

Solution

- $z_{0.01} = -2.326$:
 $\text{VaR}_{1\%} = -(0.05 - 1.4 \cdot 2.326) = 3.207\%$ (32 069 RON)
- $z_{0.025} = -1.960$, $\varphi(z_{0.025}) = 0.0584$, $\varphi/0.025 = 2.338$
- $\text{ES}_{2.5\%} = -0.05 + 1.4 \cdot 2.338 = 3.223\%$ (32 229 RON)
- Under the Normal distribution the two are almost equal (ratio 1.005)
 - ES looks inside the tail and is subadditive; VaR only marks its entrance (Chapter 7)

A6. Student- t VaR 1% and ES 2.5% [Proposed]

Solution discussed in class

Task

- ▣ Model: A5
- ▣ Same μ and σ , but a Student- t with $\nu = 5$ degrees of freedom, scaled to variance σ^2
- ▣ Use $s = \sigma \sqrt{(\nu - 2)/\nu}$, ES factor $\frac{f_\nu(q)}{\alpha} \frac{\nu + q^2}{\nu - 1}$ with $q = t_\nu^{-1}(\alpha)$
- ▣ Compare with A5: which number changes more, VaR or ES?

A7. Kupiec Test and the Basel Traffic Light [Solved]

Task

- A bank reports $x = 6$ losses above its VaR 1% in $T = 250$ days
- Kupiec coverage test [Kupiec \(1995\)](#):
 $LR_{uc} = -2[\ell(0.01) - \ell(\hat{\pi})]$,
 $\ell(p) = (T - x) \ln(1 - p) + x \ln p$
- Basel zone: green for 0–4 breaches, yellow for 5–9, red for 10 or more

Solution

- $\hat{\pi} = 6/250 = 2.4\%$; $\ell(0.01) = -30.083$, $\ell(\hat{\pi}) = -28.306$
- $LR_{uc} = 3.555$, $p = 0.059$ (χ^2_1): coverage not rejected at 5%, rejected at 10%
- $P(X \leq 4) = 0.8922$, $P(X \leq 6) = 0.9863$ under $X \sim \text{Bin}(250, 0.01)$: **yellow zone**
- The test does not reject, yet the zone is already yellow: with 250 days the test has little power, so Basel adds a capital multiplier as a precaution

A8. Kupiec over Two Years [Proposed]

Solution discussed in class

Task

- ▣ Model: A7
- ▣ The same model records $x = 12$ breaches of VaR 1% in $T = 500$ days
- ▣ Compute the breach rate, LR_{uc} and its p -value; is the model accepted?
- ▣ What does the Kupiec test not check? Which test from Chapter 8 would you add?

A9. How Many False Discoveries? [Solved]

Task

- ▣ A team tests 20 useless trading strategies, each at the 5% level, independently
- ▣ Expected number of “significant” strategies and the probability of at least one
- ▣ Bonferroni threshold for the p -value and for a two-sided $|t|$ statistic

Solution

- ▣ Expected false discoveries: $20 \times 0.05 = 1$
- ▣ $P(\geq 1) = 1 - 0.95^{20} = 0.642$
- ▣ Bonferroni: $p < 0.05/20 = 0.0025$, i.e. $|t| > 3.02$
- ▣ This is the logic behind $|t| > 3$ in the factor literature Harvey, Liu & Zhu (2016)

A10. One Hundred Factors [Proposed]

Solution discussed in class

Task

- ▣ Model: A9
- ▣ 100 candidate factors, none with a true premium, each tested at 5%
- ▣ Expected false discoveries, probability of at least one, Bonferroni $|t|$ hurdle
- ▣ Why is the Bonferroni hurdle conservative when factors are correlated?

Data for Part B

- ▣ BET index, official daily closing values, 6 Jan 2000 – 18 Sep 2026: 6 670 daily log returns in %
- ▣ Bitcoin in USD, every calendar day, 18 Sep 2014 – 18 Sep 2026: 4 384 daily log returns
- ▣ SPY (S&P 500 ETF, dividends reinvested) for B8, weekdays only
- ▣ One-day VaR 1% forecasts: window of 1000 days for GARCH- t and FHS (re-estimated every 21 days), 500 days for HS and Normal
- ▣ All code: the Quantlet `MFM_ch19_seminar` and the seminar notebook

B1. Stylised Facts of BET with Robust Inference [Solved]

Task

- ▣ Skewness, excess kurtosis and a 95% CI for the kurtosis by moving-block bootstrap (blocks of 20 days) [Künsch \(1989\)](#)
- ▣ Jarque–Bera test; Ljung–Box $Q(10)$ on r_t and on r_t^2
- ▣ Days beyond ± 4 standard deviations, against the number expected under the Normal distribution

Interpretation

- ▣ Why does the i.i.d. bootstrap understate the uncertainty of the kurtosis?
- ▣ Is the autocorrelation of r_t a sign of predictability, or of illiquidity?

B1. Solution

- Skewness -0.65 , excess kurtosis 11.1 , block-bootstrap 95% CI $[8.0, 13.9]$
- Jarque–Bera 34488: normality rejected by far
- $Q(10)$ on r_t : 108.0 ($p < 0.001$); on r_t^2 : 2440 ($p < 0.001$); first-order autocorrelation of r_t 0.109 , of $|r_t|$ 0.40
- 51 days beyond ± 4 standard deviations, against 0.42 expected under the Normal distribution
- Interpretation
 - ▶ Clustering makes neighbouring days similar; resampling blocks keeps it, resampling days destroys it
 - ▶ Positive autocorrelation of an index of less liquid stocks mostly reflects stale prices (Chapters 2, 10), not an exploitable signal

B2. Stylised Facts of Bitcoin [Proposed]

Solution discussed in class

Task

- Model: B1
- Repeat B1 for Bitcoin, 2014–2026
- Compare the width of the kurtosis CI with the BET's

Interpretation

- Which stylised facts are stronger for Bitcoin, which weaker?
- Why is the kurtosis of Bitcoin so uncertain although the sample has thousands of days?

B3. GARCH(1,1)- t for BET, with Uncertainty [Solved]

Task

- ▣ Estimate a GARCH(1,1) with Student- t innovations on BET, 2000–2026; robust standard errors
- ▣ Persistence $\alpha + \beta$ with a 95% CI by the delta method: $\text{se}(\hat{\alpha} + \hat{\beta}) = \sqrt{1^\top \hat{\Sigma}_{\alpha\beta} 1}$
- ▣ Translate the CI into a range for the half-life

Interpretation

- ▣ How long does a volatility shock on the BET last, and how sure are we?
- ▣ What does $\hat{\nu} \approx 5$ mean for VaR?

B3. Solution

- $\hat{\alpha} = 0.192$ (0.025), $\hat{\beta} = 0.799$ (0.024), $\hat{\nu} = 5.01$ (0.31)
- $\hat{\alpha} + \hat{\beta} = 0.9907$ (s.e. 0.0084); 95% CI [0.974, 1.007]
- Half-life 74 days; lower end of the CI gives 26 days, the upper end is above 1, so the half-life is unbounded
- Interpretation
 - ▶ Shocks last months, but the data cannot tell 1 month from several years: report the interval, not only 74 days
 - ▶ $\hat{\nu} \approx 5$: even conditionally, tails are heavy; a Normal GARCH VaR 1% would be too low

B4. Asymmetry: GJR-GARCH- t for BET [Proposed]

Solution discussed in class

Task

- Model: B3
- Add a leverage term:
$$\sigma_t^2 = \omega + (\alpha + \gamma \mathbb{I}\{\varepsilon_{t-1} < 0\})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$
- Test $\gamma = 0$ with the t statistic and a likelihood-ratio test; compare AIC and BIC

Interpretation

- Do the tests and the criteria agree? Which model would you use for VaR?

B5. Backtesting VaR 1% for BET, 2005–2026 [Solved]

Task

- ▣ Four one-day VaR 1% forecasts: HS, Normal, GARCH- t , FHS McNeil & Frey (2000)
- ▣ For each: number and rate of breaches, Kupiec p_{uc} , Christoffersen p_{ind} , joint p_{cc} Kupiec (1995), Christoffersen (1998)
- ▣ Share of 250-day windows in the Basel red zone

Interpretation

- ▣ Which model would a regulator accept? Which one would a trader prefer, and why might they differ?

B5. Solution

Model	Breaches	Rate	p_{uc}	p_{ind}	p_{cc}	Red zone
HS	79	1.45%	0.002	<0.001	<0.001	5.5%
Normal	122	2.25%	<0.001	<0.001	<0.001	18.6%
GARCH-t	80	1.47%	0.001	0.864	0.005	0.0%
FHS	67	1.23%	0.095	0.267	0.134	0.0%

- $T = 5\,432$ days, about 54 breaches expected
- Interpretation
 - ▶ Only FHS passes all three tests; HS and Normal breach in clusters, GARCH-t too often
 - ▶ A regulator wants correct coverage; a trader wants low capital: the Normal model has the lowest average VaR (2.90% vs FHS 3.17%) and fails

B6. Backtesting VaR 1% for Bitcoin, 2019–2026 [Proposed]

Solution discussed in class

Task

- Model: B5
- Repeat B5 for Bitcoin, evaluation from 2019
- Plot losses and the HS and FHS forecasts since 2022

Interpretation

- Is the ranking of models the same as for BET?
What changes when the market trades every day?

B7. The Look-Ahead Trap [Solved]

Task

- Recompute the BET VaR 1% of B5 with parameters estimated on the **full** sample 2000–2026
- Two versions
 - ▶ HS: one quantile of all losses 2005–2026
 - ▶ GARCH- t : full-sample parameters, $\hat{\sigma}_t$ still filtered day by day
- Compare the breach rates with the rolling versions

Interpretation

- Why does the look-ahead version look better?
Could anyone have used it in 2008?

B7. Solution

- Full-sample HS: VaR 4.24% every day, 55 breaches (1.01%): exactly 1% by construction
- Full-sample GARCH- t : 71 breaches (1.31%, p_{uc} 0.030); rolling: 80 (1.47%)
- Interpretation
 - ▶ The full sample includes 2008, 2020 and every later crash: the model has already seen them
 - ▶ No one could have used these parameters in real time; only the rolling result is evidence

B8. The Calendar Trap: Bitcoin and SPY [Proposed]

Solution discussed in class

Task

- Model: B7
- Compute Bitcoin–SPY statistics in two ways
 - ▶ Wrong: daily returns on each own calendar, then keep the common days
 - ▶ Right: keep the common days of the prices, then compute returns
- 2015–2019 and 2020–2026: correlation, Monday standard deviation of Bitcoin, cumulative Bitcoin return

Interpretation

- Which statistic is most damaged by the wrong join?
Would a portfolio VaR be too low or too high?

Proposal Template (One Page)

Sections

- ▣ **Title and team**
- ▣ **Question:** one sentence, ending with a question mark
- ▣ **Hypothesis and benchmark:** H_0 , and the simple model you must beat
- ▣ **Data:** series, source, period, frequency, cleaning rules

Sections (continued)

- ▣ **Methods:** models, estimation window, test period, loss function, tests
- ▣ **Expected outcome:** what result would change your mind?
- ▣ **Risks:** data problems, too few events, overfitting
- ▣ **References:** three to five, clickable, each checked to exist (DOI or link)

A Worked Proposal

- **Question:** did the tail risk of the BET change after the FTSE reclassification of the BVB as a Secondary Emerging market (21 September 2020)?
- **Hypothesis:** H_0 : same excess kurtosis, GARCH persistence and VaR 1% breach rates in the six years before and after
- **Data:** BET official closes, 22 Sep 2014 – 18 Sep 2026; holiday fills removed
- **Methods:** block-bootstrap CI for the kurtosis difference; GARCH(1,1)- t in each period; rolling HS and FHS VaR 1%, Kupiec tests by period
 - ▶ **Benchmark:** no change; **control:** the same statistics for a market without the event (for example the WIG20)
- **Risks:** the pre-period contains the COVID-19 crash; one event, so no causal claim without the control market

C1. Your Team's Proposal [Proposed]

- ▣ Model: the worked proposal above; seeds: Part C of Seminars 0–18
- ▣ Fill in the template in one page and submit it within one week
- ▣ Checklist before you submit
 - ▶ Which published number will you replicate first (step 1 of the project)?
 - ▶ Is the question answerable with data you already have?
 - ▶ Is the benchmark strong, and is the loss function fixed in advance?
 - ▶ Does every forecast use only information available at $t - 1$?
 - ▶ Have you counted how many models and tests you will run?

C2. Critique an AI-Written Backtest (1/2) [Proposed]

Solution discussed in class

- ▣ **Student prompt:** “bet holds the daily BET closes. Write a backtest of historical-simulation VaR 1% with a 500-day window and a Kupiec test, and give me a reference for the result.”
- ▣ **Answer of the AI assistant**

```
import numpy as np
from scipy.stats import chi2
r = np.log(bet).diff().dropna()
var = -r.rolling(500).quantile(0.01)
hit = (-r > var)[var.notna()]
x, n, p = hit.sum(), hit.size, 0.01
pi = x / n
lr = -2 * (x*np.log(p) + (n-x)*np.log(1-p) - x*np.log(pi) - (n-x)*np.log(1-pi))
print(x, n, chi2.sf(lr, df=2))
```

- ▣ “The breach rate is close to 1% and the Kupiec p -value is above 5%: historical simulation is unbiased for BET, as shown by Kupiec and Christoffersen (2004), *Backtesting historical simulation VaR*, Journal of Risk Model Validation, 8(2), 45–67.”

C2. Critique an AI-Written Backtest (2/2) [Proposed]

Solution discussed in class

- The answer on the previous slide contains **three** errors that often appear in AI-written project code; the rest is correct
 - ▶ Model: B5 and B7 (backtest and the look-ahead trap), A7 (Kupiec test)
- Your tasks
 - ▶ Find each error: which line or sentence, and why it is wrong
 - ▶ Say how you would detect it without knowing the answer: a timing test, the number of restrictions, a DOI search
 - ▶ Give the corrected code and a correct reference
 - ▶ Log each error as in [AI_ERRORS.md](#): what the AI answered, what was wrong, how you found out, the correct version

Common Pitfalls in Proposals

- ▣ A topic instead of a question (“Bitcoin volatility”)
- ▣ No benchmark, or no test period fixed in advance
- ▣ Data that do not exist at the needed frequency or history
- ▣ Too many markets and models for the time available
- ▣ A single event treated as a causal experiment without a control

Take-Home Assignment (Team Project)

- Submit your one-page proposal (C1) within one week
- Step 1 of the project: choose the published number you will replicate and give its source with a DOI
- Open the team repository on GitHub with a README and the two mandatory files
 - ▶ [AI_USE.md](#): which AI tools you used, for what, and what you checked yourselves
 - ▶ [AI_ERRORS.md](#): at least three AI errors you caught
- Log any AI error you catch from now on in [AI_ERRORS.md](#), as in C2

Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Statistics of Financial Markets
- ☐  Measuring Statistical Risk

Code and data

- ☐ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_19

References

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